

SHIJIAZHUANG ZHENGDING INTL, China

WMO: 470312

Lat: 38.281N

Lon: 114.697E

Elev: 233

StdP: 14.57

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.8	14.1	-18.0	2.1	29.5	-12.7	2.8	29.7	21.3	28.1	17.3	31.5	3.6	320	0.430

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	97.1	72.1	94.7	73.2	91.5	73.5	81.7	88.6	79.9	86.3	78.4	84.8	8.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.3	158.9	85.9	78.5	149.4	84.5	76.7	140.6	82.9	45.7	89.4	43.7	87.1	42.0	85.6	89.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.4	14.3	12.0	DB	7.0	102.7	3.3	3.6	4.7	105.2	2.8	107.3	1.0	109.3	-1.4	111.9	
			WB	5.2	84.9	2.9	2.2	3.2	86.5	1.5	87.8	-0.1	89.1	-2.2	90.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.8	26.2	32.5	45.7	57.9	68.9	78.1	81.1	77.8	69.4	58.0	42.0	29.9
	DBStd	19.94	4.85	6.08	7.58	6.60	5.69	5.21	4.11	4.33	5.14	6.11	7.52	5.23
	HDD50	2301	737	489	176	13	0	0	0	0	0	7	257	622
	HDD65	4988	1202	909	598	226	26	0	0	0	21	229	690	1087
	CDD50	4404	0	0	44	250	586	844	963	863	583	255	16	0
	CDD65	1618	0	0	1	13	148	395	498	398	154	11	0	0
	CDH74	16701	0	0	35	265	1664	4429	5374	3577	1189	168	0	0
	CDH80	7244	0	0	7	60	612	2274	2524	1397	342	28	0	0
(14) Wind	WSAvg	5.0	4.5	5.0	6.3	6.6	6.0	5.7	4.7	3.9	4.0	4.3	4.5	4.7
(15) Precipitation	PrecAvg	21.7	0.2	0.3	0.5	1.0	1.5	2.7	5.9	5.2	2.5	1.0	0.7	0.2
	PrecMax	36.1	0.7	1.4	1.9	2.2	3.1	5.2	14.3	16.3	4.3	3.1	2.4	0.8
	PrecMin	12.6	0.0	0.0	0.0	0.2	0.4	1.0	2.5	1.7	0.4	0.2	0.0	0.0
	PrecStd	4.9	0.2	0.3	0.6	0.6	0.7	1.3	2.8	3.0	1.3	0.7	0.7	0.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.9	60.6	78.4	86.3	95.3	102.3	101.8	95.2	91.0	83.8	69.9	53.4
		MCWB	35.4	42.0	54.5	65.6	66.2	69.3	72.3	80.2	68.5	63.2	50.8	38.9
	2%	DB	44.5	54.0	71.7	80.7	89.9	98.3	97.1	91.6	86.4	78.7	64.1	48.5
		MCWB	32.2	39.0	52.1	61.5	66.6	68.9	75.5	78.3	68.8	59.7	49.6	35.5
	5%	DB	40.7	49.6	66.2	76.8	86.2	95.0	94.7	89.6	84.1	75.0	58.8	44.4
		MCWB	30.5	37.3	49.1	58.8	66.0	69.3	76.4	76.9	67.6	57.3	46.3	33.9
	10%	DB	37.1	44.7	60.9	73.0	82.7	91.7	91.4	87.6	80.8	70.1	55.0	40.6
		MCWB	28.7	34.4	46.0	57.6	64.4	69.7	76.2	76.2	67.3	56.3	45.5	31.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.9	44.0	59.4	69.3	74.9	76.8	84.4	84.2	74.8	67.1	54.2	40.1
		MCDB	47.2	56.1	69.7	81.1	86.3	90.9	93.1	89.8	83.1	75.3	63.3	48.4
	2%	WB	33.6	40.3	54.6	65.8	72.3	75.1	82.2	81.8	72.8	63.6	51.6	37.0
		MCDB	41.2	51.8	68.9	76.0	83.4	88.4	90.1	87.7	80.1	71.9	58.5	46.5
	5%	WB	31.3	37.8	50.8	62.4	70.0	73.6	80.4	80.2	71.2	61.3	49.2	34.7
		MCDB	39.1	47.9	63.7	71.5	80.3	86.5	87.5	85.8	78.3	68.7	56.8	42.8
	10%	WB	29.2	35.1	47.1	59.1	67.5	72.2	78.9	78.7	69.7	59.4	46.6	32.6
		MCDB	35.7	43.6	59.0	69.0	77.5	84.5	85.9	84.0	77.0	66.0	52.8	38.6
(35) Mean Daily Temperature Range	5% DB	MDBR	20.3	20.7	24.1	23.8	23.9	22.5	17.1	16.1	20.0	21.2	20.0	19.5
		MCDBR	26.6	29.0	32.8	31.4	30.6	29.4	23.3	19.7	25.3	27.7	26.1	26.0
		MCWBR	17.4	17.3	17.2	16.3	14.3	8.9	8.1	9.0	10.4	12.0	14.7	16.2
	5% WB	MCDBR	23.9	25.8	30.2	27.4	25.6	22.4	17.6	15.9	18.9	20.7	22.2	23.0
		MCWBR	16.3	16.2	17.0	16.7	14.5	9.2	8.4	8.4	9.7	10.7	13.4	15.0
(40) Clear-Sky Solar Irradiance	taub		0.480	0.557	0.559	0.611	0.651	0.794	0.790	0.770	0.699	0.642	0.518	0.472
	taud		1.836	1.701	1.725	1.666	1.627	1.442	1.489	1.504	1.578	1.632	1.815	1.890
	Ebn at Noon		202	202	222	220	215	186	185	183	185	177	188	193
	Edh at Noon		49	64	68	76	81	98	93	89	78	68	49	43
(44) All-Sky Solar Radiation	RadAvg		794	993	1436	1712	1885	1706	1507	1455	1292	1044	822	722
	RadStd		63	123	128	111	147	157	98	122	108	114	130	76

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page